V.I.S.O.R. Project

Systems Engineering SAT Part A: Investigative Report

Aryan Gupta

Student ID: 000000000X

April 30, 2026

"Design is not just what it looks like and feels like. Design is how it works."
— **Steve Jobs**

Contents

Acknowledgements

I would like to express my sincere gratitude to my Systems Engineering teacher for their invaluable guidance and technical feedback throughout the initial research and design phase of this project. Their expertise helped me refine the scope of V.I.S.O.R. and navigate the complexities of the systems engineering process.

I am also grateful to the laboratory technicians and staff at The Geelong College for providing access to the necessary tools and facilities required to investigate my design options. Finally, I would like to thank my peers for their constructive criticism and support during our collaborative brainstorming sessions.

Project Design & Plan Overview

Project Overview - Problem Statement & Objectives

The purpose of this project is to assist students and staff in school technology workshops who face the challenge of the ”Panic Gap”–a critical delay in first aid treatment caused by shock, hesitation, or a lack of medical knowledge during minor emergencies. The key objective of this project is to build an integrated and controlled mechanical and electrotechnological system, titled V.I.S.O.R. (Visual Inspection & Smart Occupational Relief), that autonomously diagnoses an injury and dispenses the correct treatment.

The system must meet the following key objectives and evaluation criteria:

- Dispensing Reliability: Achieve a > 90% success rate without the flexible medical packaging jamming.
- Response Latency: Process user input and dispense the item in < 15 seconds to effectively bridge the Panic Gap.
- Interpretation Accuracy: Achieve a 100% logical match between the user's distress signal and the dispensed item using AI.
- Integration: Successfully utilize a dual-microcontroller architecture (Raspberry Pi 4 and Arduino Uno).

Key Learnings from Research and Inspiration

Research was conducted using commercial first aid kits, spiral vending machines, and smart pill dispensers. It was found that:

1. Passive first aid kits do nothing to alleviate the cognitive load on a panicked user.
2. Standard gravity-fed or spiral dispensing mechanisms are highly prone to friction jams when handling soft, flexible items like bandage wrappers.
3. While offline voice-recognition is excellent for privacy, environmental testing revealed that ambient workshop noise (often exceeding 80+ decibels) renders acoustic sensors entirely unreliable. Therefore, a visual/camera-based AI approach is required for reliable input.

Preferred Design Option - Outline & Justification

Potential solutions for the end-user included an Acoustic-Triggered Rotary Carousel, an RFID-Authenticated Solenoid Gravity Gate, and an AI-Driven Rack and Pinion Dispenser.

The solution decided upon was the AI-Driven Rack and Pinion Dispenser (V.I.S.O.R.) for the following reasons:

1. **Mechanical Reliability**: Unlike the rejected options which relied on gravity and caused soft packaging to snag, the rack and pinion mechanism driven by a continuous rotation servo applies active, linear horizontal thrust. This forcibly pushes the item out of the cartridge, guaranteeing the > 90% reliability criteria.
2. **Elimination of the Panic Gap**: By utilizing the Gemini AI API via a Raspberry Pi 4 camera, the user is not forced to navigate menus, find an ID card, or know what medical supply they need. The AI acts as the triage nurse, fulfilling the core requirement of rapid, stress-free treatment under the 15-second latency threshold.
3. **Optimal Subsystem Integration**: Delegating high-level AI API processing to the Raspberry Pi and low-level hardware PWM timing to the Arduino Uno creates a highly stable, complex electromechanical system suitable for VCE Systems Engineering standards.

Preferred Design Option - Diagrammatic Overview

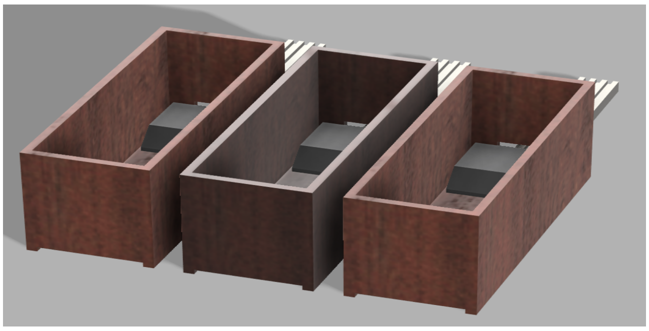


Figure 1: *Basic CAD render of V.I.S.O.R.*

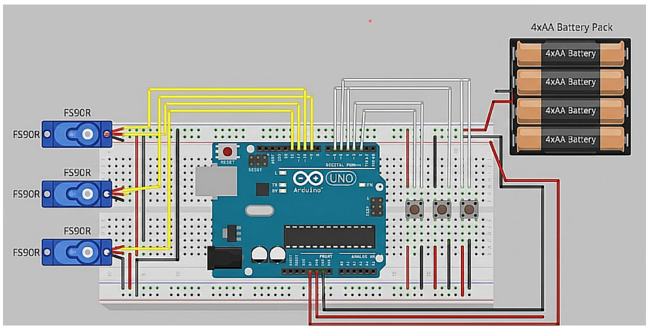


Figure 2: *Schematic diagram of V.I.S.O.R. cartridges*

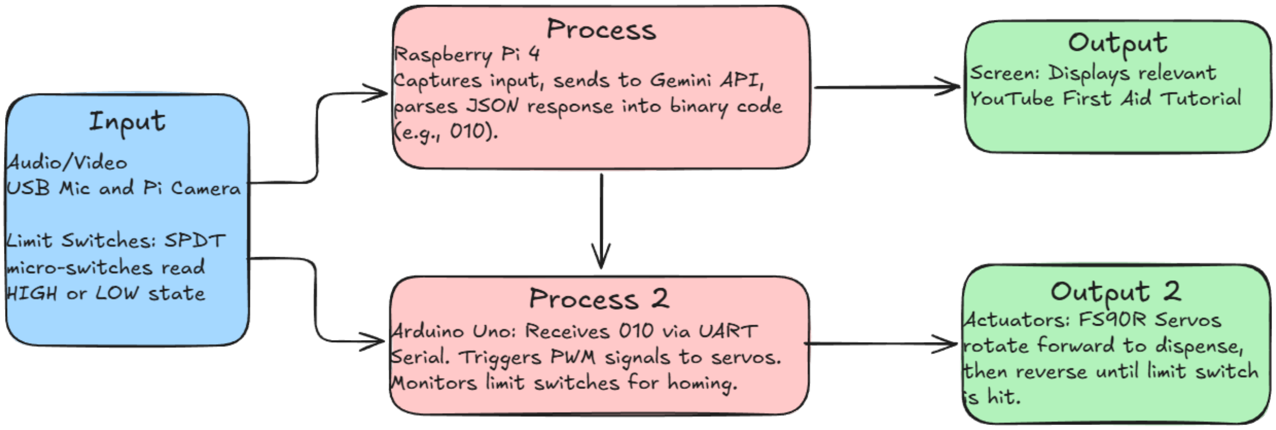


Figure 3: *Input-Process-Output (IPO) model for V.I.S.O.R.*

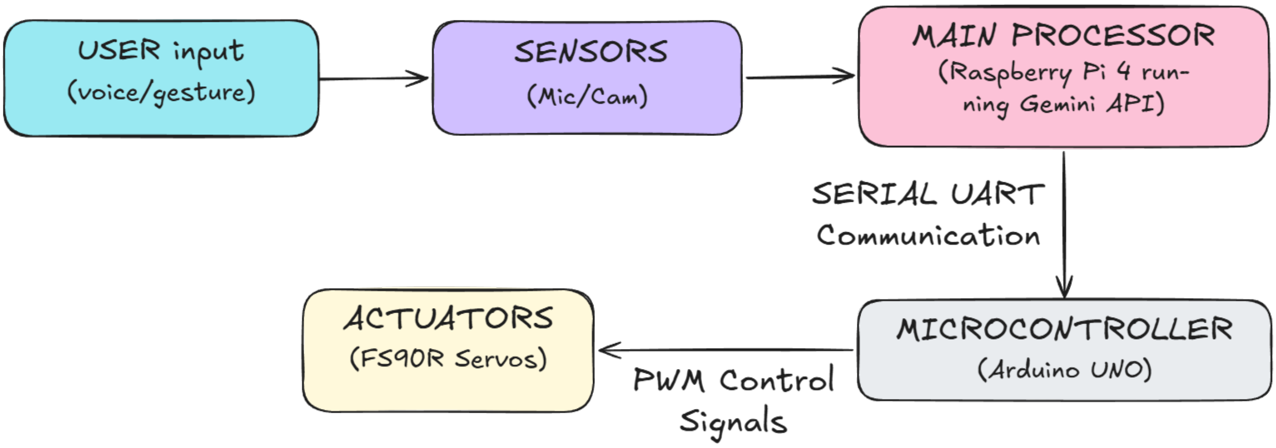


Figure 4: *Integrated System Flow Diagram for V.I.S.O.R.*

1 Design Brief

The "Visual Inspection & Smart Occupational Relief" (V.I.S.O.R) system is an automated first aid console designed to address the critical "Panic Gap" often experienced during workshop injuries. When an accident occurs, untrained individuals often hesitate or misapply treatment due to shock or lack of knowledge. V.I.S.O.R aims to eliminate this latency by utilising the Gemini AI API to interpret user distress signals (via voice or camera) and autonomously dispensing the correct medical supply (Bandages, Gauze or Alcohol Pads) while simultaneously displaying a relevant instructional video.

This project is constrained by a student budget of $\$100$ and a strict Year 12 timeline. It requires the integration of complex subsystems, specifically a dual-microcontroller architecture where a Raspberry Pi 4 handles high-level AI processing and computer vision, communicating via serial connection to an Arduino Uno which manages low=level electromechanical actuation. The design priorities user safety, mechanical reliability, and data privacy, ensuring that no biometric data is permanently stored.

The successful V.I.S.O.R prototype will be self-contained, mains-powered unit capable of identifying at least three distinct injury types with 100% logical accuracy and dispensing the appropriate relief within 15 seconds, thereby reducing the cognitive load on the injured user and ensuring safe, effective first aid treatment.

1.1 Context & Problem Statement

In high-risk environments such as school technology workshops or industrial makerspaces, minor injuries (cuts, abrasions, burns) are common. While First Aid kits are legally required in these spaces, they are passive systems. When an inexperienced student or worker suffers an injury, they often face the "Panic Gap"—a period of hesitation caused by shock or a lack of medical knowledge. They may not know which supply to use, or how to apply it correctly. This delay can lead to complications or increased distress for the injured individual.

Figure 1.1: *The 'Panic Gap' represents the critical delay between injury and treatment.*

Figure 1.2: *Standard passive first aid kits require the user to hunt for supplies, increasing stress during an emergency.*

The problem is that current first aid solutions rely entirely on the user's pre-existing knowledge. In a moment of stress, a user may struggle to identify the correct procedure, leading to improper treatment or infection. There is a need for an automated, intelligent system that can bridge this gap by diagnosing the need via natural language or visual input and immediately providing both the physical supplies and the instructional guidance required to treat the injury effectively.

1.2 Constraints

- **Financial:** The project must be completed within the budget of approximately $\$100$ AUD.
- **Time:** The design, construction, and testing must be completed within the Year 12 Systems Engineering timeline (Terms 1–3).
- **Physical Size:** The unit must be desktop-portable and self-contained, not exceeding dimensions of $400mm \times 300mm \times 300mm$.

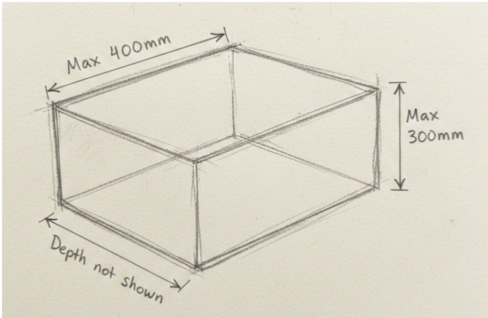


Figure 1.3: *Maximum physical dimensions for the desktop prototype*

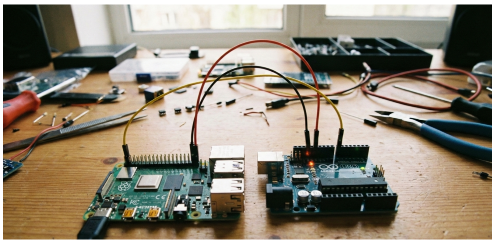


Figure 1.4: *Dual-microcontroller setup (Pi for AI, Arduino for Motors)*

- **Legal/Safety:** The system cannot dispense oral medication (e.g., painkillers) due to legal restrictions. It is limited to topical supplies.
- **Technical:** The system must utilize a dual-microcontroller architecture to demonstrate complex system integration.

1.3 Requirements

- **Input System:** The system must accept user input via audio (microphone) or visual data (camera) to assess the situation.
- **Processing:** It must utilise the Gemini AI API to interpret the user's distress signal and generate a specific dispensing command (Binary $0|1$ format).
- **Output (Physical):** It must be capable of dispensing at least three distinct types of first aid supplies (Bandages, Gauze, Alcohol Pads) using electromechanical actuators.
- **Output (Visual):** It must display a relevant YouTube instructional video or visual guide corresponding to the dispensed item.

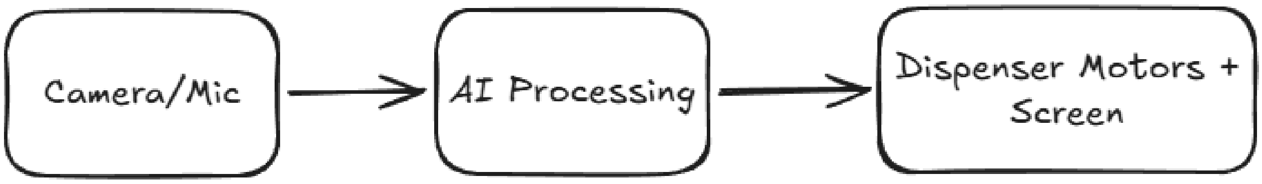


Figure 1.5: *High-level requirements flow showing the transition from AI interpretation to physical actuation.*

Figure 1.6: *The three specific consumables the system is required to dispense*

- **Maintenance:** The cartridges must be refillable, and the system must allow for maintenance access without destroying the chassis (e.g., via a rear service door).

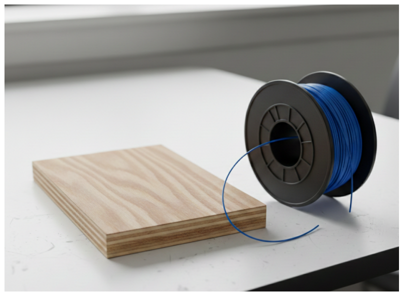
1.4 Considerations

Figure 1.7: *Data privacy is a key consideration; user video feeds are processed locally or transiently and never stored.*

- **Privacy:** Since the device uses a camera and microphone, user data (images/audio) must be processed locally or via secure API calls and not stored permanently, ensuring user privacy is maintained.
- **Hygiene:** The dispensing mechanism must be designed to minimize contact with the sterile parts of the first aid supplies. The “pusher” mechanism should only touch the external packaging.
- **Power Redundancy:** In a workshop blackout, the device would lose power. A consideration is whether a backup battery is feasible, though mains power is the primary standard for this prototype.

1.5 System Design & Implementation Drivers

Table 1.1: *Analysis of Influencing Factors on the V.I.S.O.R. Development Lifecycle*

Factor	Relevance to V.I.S.O.R
User Safety	High Priority. As a medical assistance device, the system itself must not cause harm. The chassis must have no sharp edges (sanded plywood/filleted $3D$ prints), and the $240V$-to-$5V$ power supply must be electrically isolated and enclosed to prevent shock hazards in a workshop environment.
Function/Reliability	Critical. If the user cuts their hand, the machine must dispense the bandage. Jamming is not acceptable. The rack-and-pinion tolerances must be tuned to ensure a > 90% dispensing success rate.
Sustainability	The chassis will be constructed from plywood (a renewable resource) rather than acrylic. 3D printed components will use PLA (Polylactic Acid), a bioplastic derived from cornstarch, to minimize the long-term environmental impact of the prototype.  **Figure 1.8:** *Selection of PLA bioplastic and plywood ensure the project meets sustainability factors.*
User Needs	The screen must be angled for visibility by a standing or seated user. The interface must be "hands-free" or require minimal touch, as the user's hands may be injured or dirty.

1.6 Evaluation Criteria

Table 1.2: *Quantitative and Qualitative Evaluation Criteria for System Validation*

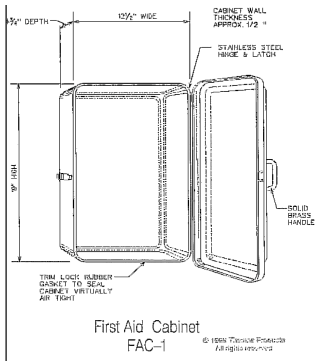
Criteria	Type & Method of Assessment	Expected Value / Range	Relation to Requirements	Relation to Influencing Factors	Justification of Expected Value
Dispensing Reliability	Quantitative. Run the "dispense" command 20 times for each cartridge type (Bandage, Gauze, Wipe). Record pass/fail.	$>$ 90% Success Rate (18/20 Successful Dispenses)	The brief requires the system to dispense three distinct types of supplies autonomously. A failure here means the core function is not met.	Relates to Function/Reliability. In an emergency, a jamming mechanism increases the "Panic Gap" rather than reducing it.	While 100% is ideal, a 90% success rate allows for minor friction inconsistencies in $3D$ printed parts while still being reliable enough for a prototype.
AI Response Latency  **Figure 1.9:** *The 15-second latency threshold is the primary quantitative metric for success*	Quantitative. Measure the time from the end of the user's spoken sentence to the moment the video appears and the motor starts.	$<$ 15 Seconds	To be useful in a "Panic Gap" situation, the system must be faster than finding a manual kit and reading a booklet.	Relates to User Needs (Ergonomics). An injured user is stressed and impatient; the system must feel responsive and helpful immediately.	15 seconds is the estimated threshold of user patience before they panic or give up. Any longer renders the "smart" aspect hindrance.
Interpretation Accuracy	Qualitative. Test with 10 different phrases (e.g., "I cut my finger", "I have a burn"). Assess if the AI selects the logically correct item.	100% Logical Match	The system must distinguish between a cut (Bandage) and a burn (Gauze/Cooling). Incorrect dispensing renders the device useless.	Relates to User Safety. Incorrect advice or supplies (e.g., putting alcohol on a serious burn) could worsen the injury.	There is zero margin for error in medical advice. Even one incorrect diagnosis makes the system unsafe to use.

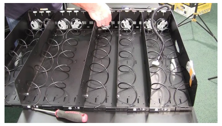
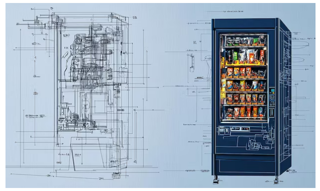
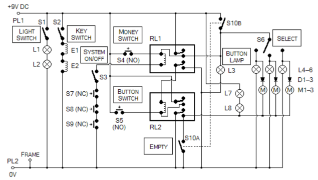
Criteria	Type & Method of Assessment	Expected Value / Range	Relation to Requirements	Relation to Influencing Factors	Justification of Expected Value
Mechanical Safety	Qualitative. Visual inspection of the case and internal wiring. Check for exposed wires or sharp edges.	Zero hazards found.	Relates to the "User Safety" factor. A safety device cannot introduce new hazards to the user.	Relates to User Safety and Legal/Standards. A first aid device must not cause further injury to the user.	Safety is a binary pass/-fail. Any exposed wire or sharp edge is an automatic fail in a school environment.
Refill Efficiency	Quantitative. Time taken to open the back door, reload a cartridge with 5 items, and close the unit.	< 2 Minutes	Ensures the "Maintenance" requirement is met, making the device practical for a school or workshop custodian to maintain.	Relates to Sustainability and Maintenance. If the device is difficult to refill, it is likely to be discarded or left empty, reducing its lifespan.	2 minutes is a reasonable timeframe for a teacher or cleaner to restock the device during a short break between classes.

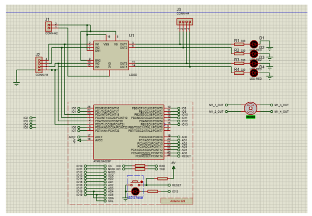
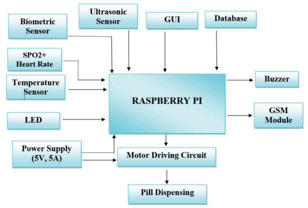
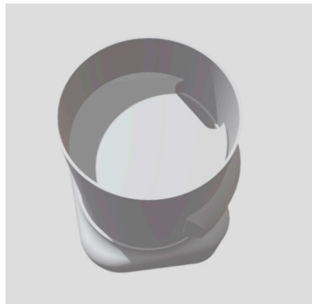
2 Research and Design

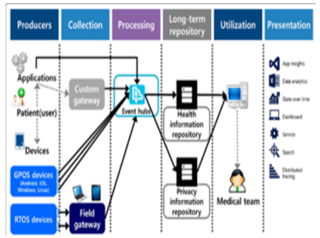
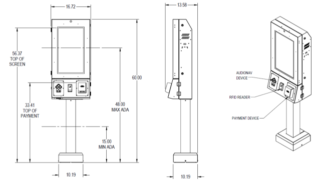
2.1 Environmental Scan & Existing Solutions

Table 2.1: *Comparative Analysis of Existing Solutions and Design Inspiration for V.I.S.O.R.*

Idea Title & Image	**Positives**	**Negatives**	**Interesting Aspects**	**Key Components/ Subsystems/ Process**	**Sketches/ Circuit Diagrams/ Technical Data**	**URL**
Standard Wall-Mounted First Aid Kit  *Figure 2.1: A traditional, passive wall-mounted first aid cabinet.*	Meets the requirement of holding Bandages, Gauze, and Wipes. It is low cost and requires no power.	It is completely passive. It relies entirely on the user's prior medical knowledge, doing nothing to solve the "Panic Gap" or provide instructions.	The standard sizing of the internal compartments provides good reference dimensions for V.I.S.O.R's $3D$ printed cartridges.	Plastic/Metal chassis, hinges, latches, sterile supplies.	 	N/A - General Consumer Product.

Idea Title & Image	Positives	Negatives	Interesting Aspects	Key Components/ Subsystems/ Process	Sketches/ Circuit Diagrams/ Technical Data	URL
Coil-Based Vending Machine  *Figure 2.2: Standard spiral vending machine mechanisms, prone to jamming with flexible packaging.*	Fully automated dispensing. Highly reliable for rigid items (like boxes or cans). Utilizes a simple rotational-to-linear physical mechanism.	Extremely bulky and expensive. Furthermore, spiral coils are notorious for jamming when trying to dispense flat, flexible items like individual bandages.	The use of a glass front allows users to see inventory levels—a feature worth replicating or mimicking.	Keypad input, Microcontroller, Stepper Motors, Spiral metal coils, drop-chute.	 	Instructables Link

Idea Title & Image	Positives	Negatives	Interesting Aspects	Key Components/ Subsystems/ Process	Sketches/ Circuit Diagrams/ Technical Data	URL
Smart Pill Dispenser  *Figure 2.3: A commercial smart pill dispenser, utilizing excellent UI but limited to rigid pills.*	Solves the “cognitive load” issue by giving the user exactly what they need automatically. Uses a sleek, medical-grade UI and app connectivity.	It operates on a timer (schedule) rather than reacting to an emergency physical input. It uses a rotary drum designed only for tiny, rigid pills, not flat wound-care supplies.	The aesthetic design is highly ergonomic and approachable, reducing user anxiety.	Wi-Fi module, Rotary drum carousel, DC motors, LCD interface, App integration.	  	Instructables Link

Idea Title & Image	Positives	Negatives	Interesting Aspects	Key Components/ Subsystems/ Process	Sketches/ Circuit Diagrams/ Technical Data	URL
Telehealth Kiosk / Smart Mirror  *Figure 2.4: A telehealth kiosk demonstrating the effectiveness of visual/audio triage interfaces.*	Uses cameras and AI/remote doctors to diagnose users in real-time. Provides exact, tailored advice to the user's specific problem.	Does not physically dispense immediate relief items. Usually costs thousands of dollars, completely violating the $\$100 - 150$ budget constraint.	Proves that visual/audio triage is highly effective for rapid medical assessment.	High-definition camera, Touchscreen monitor, broadband internet connection, cloud processing.	 	N/A - Commercial Enterprise System

2.2 Literature Review

2.2.1 Foundational Engineering Principles

- **Cloud Based Artificial Intelligence (LLM/VLM APIs)**: The system relies on Large Language Models and Vision-Language Models (specifically the Gemini API) to process unstructured natural language or visual data (e.g., "I cut my finger") to output structured deterministic machine data (e.g., a binary 010). This bridges the gap between human panic and machine logic.
- **Asynchronous Serial Communication (UART)**: Because the system utilizes a dual-microcontroller architecture, data must be passed between the Raspberry Pi and the Arduino. Universal Asynchronous Receiver-Transmitter (UART) serial communication relies on agreed-upon baud rates (e.g., 9600 bps) to transmit the AI's binary dispensing commands as electrical pulses over TX/RX lines.
- **Kinematics (Rack and Pinion)**: A core engineering challenge is dispensing flat, flexible items (bandages) that cannot be easily dropped or coiled. A rack and pinion gear system translates rotational kinetic energy (from a servo motor) into linear force. This allows a "pusher sled" to apply horizontal thrust to the back of the medical supply, sliding it linearly out of the cartridge slit.

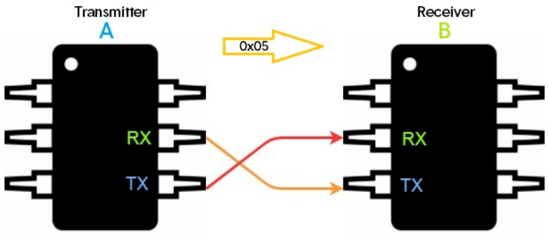


Figure 2.5: *UART Serial Communication architecture, required to link the high-level Raspberry Pi with the low-level Arduino.*

2.2.2 Components and Sub Systems

- **Raspberry Pi 4 Model B**: Serving as the high-level logic controller. It possesses the necessary RAM ($4GB$) and processing power to run a Linux OS, interface with USB microphones/cameras, handle HTTPS web requests to the Gemini API, and drive an HDMI display for the instructional YouTube videos.
- **Arduino Uno $R3$**: Serving as the low-level motor controller. While the Pi struggles with precise hardware PWM (Pulse Width Modulation) due to its operating system overhead, the Arduino excels at it. It reads the serial command from the Pi and safely generates the PWM signals required to actuate the servos.
- **Continuous Rotation Servos (e.g., $FS90R$ or $MG996R$)**: Standard servos are locked to a 180-degree sweep. Continuous rotation servos have their internal potentiometers decoupled, allowing them to spin 360 degrees continuously based on PWM signals. This is required to drive the pinion gear multiple rotations to push the rack a sufficient linear distance.
- **Micro Limit Switches**: Acting as closed-loop feedback sensors. When the rack extends fully and dispenses the item, the motor must reverse. A limit switch placed at the rear of the cartridge will physically click when the rack retracts to its "home" position, telling the Arduino to cut power to the servo.

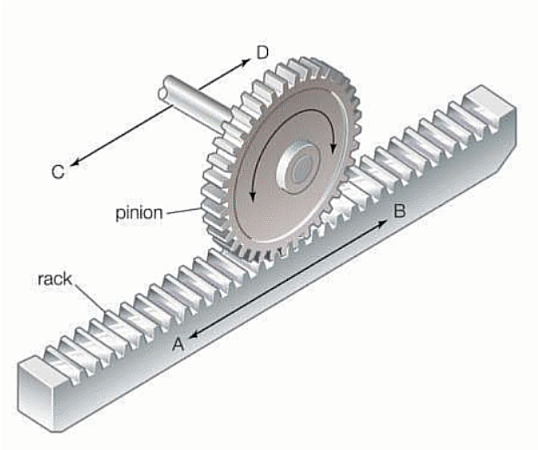


Figure 2.6: *Rack and Pinion Gear System for Linear Motion Translation.*

2.2.3 Production Processes

- **Fused Deposition Modelling (FDM) $3D$ Printing**: The internal cartridges, rack and pinion gears, and pusher sleds require tight tolerance (e.g., a $1.5mm$ sliding gap) and complex geometries that are impossible to create via manual woodworking. FDM printing using PLA filament allows for rapid prototyping and iterative testing of the dispensing slit dimensions to ensure bandages do not jam.
- **Computer Numerical Control (CNC) Routing or Laser Cutting**: The external chassis needs to safely house high-voltage ($240V$ to $5V$) power supplies, the screen, and the electronics. Plywood ($6mm$) will be utilized for its structural rigidity and sustainability. Using a CNC router or Laser cutter ensures perfectly square joints and precise cutouts for the screen bezel and dispensing tray, yielding a professional aesthetic.

2.3 Design Options

2.3.1 Design Option 1

Table 2.2: *Technical Specifications and System Description for Design Option 1*

Design Option 1 - V.I.S.O.R	
System Description	This design relies on a dual-microcontroller architecture to provide an intelligent, active-dispensing first aid console. A Raspberry Pi 4 uses a camera and microphone to capture user input, sending it to the Gemini AI API for evaluation. Once the injury is classified, the Pi sends a binary command via serial to an Arduino Uno. The Arduino then powers a continuous rotation servo attached to a $3D$-printed rack and pinion mechanism, which actively pushes the correct first aid supply out of the cartridge.
System Function/s	The system acts as a fully autonomous triage and dispensing unit. It takes visual/audio input, processes the logical diagnosis using AI, and outputs a physical item (using linear mechanical thrust) alongside an instructional video on an integrated screen.
System Components	–
Control Systems	Raspberry Pi 4 Model B (High-level AI logic and video output) and Arduino Uno $R3$ (Low-level PWM motor control).
Sensors	USB Microphone and Raspberry Pi Camera Module $V3$ for capturing the user's distress signal. Micro Limit Switches to detect when the rack has returned to its "home" position.
Actuators	$FS90R$ or $MG996R$ Continuous Rotation Servos.
Mechanical Structures	A $3D$-printed rack and pinion gear system converts the servo's rotational motion into linear thrust. A wedge-shaped "pusher sled" slides along a track to push flexible bandage packets out of a narrow $2.5mm$ slit in the front of the cartridge.
Programming Structures	Python script on the Pi handling API HTTPS requests and UART Serial Communication. C++ on the Arduino interpreting the serial bytes and controlling servo PWM signals.
User Interface	A 7-inch HDMI touchscreen displays YouTube first aid tutorials. The initial interaction is entirely natural language/voice-driven.
Power Sources	$240V$ AC to $5V$ DC ($3A+$) mains wall adapter to power the Pi, Arduino, screen, and servos simultaneously without brownouts.
System Materials / Constructions	Exterior chassis made of $6mm$ plywood for rigidity and ease of manufacturing. Internal cartridges, gears, and sleds are $3D$ printed using PLA.
Knowledge Requirements – Difficulty Levels	Level 5 (Very Advanced). Requires API integration, Python/C++ serial handshakes, and precise CAD parametric modelling for gear tolerances.

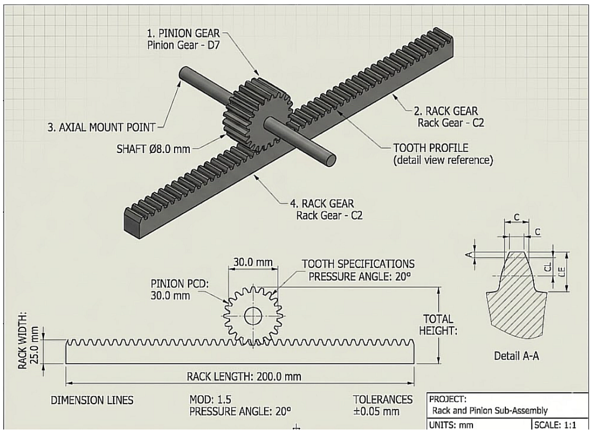
Table 2.3: *Design Option 1 Performance Ratings Against Key Requirements*

Ratings						
User Needs / Requirements	0	1	2	3	4	5
Safety	0	1	2	3	4	5
Environmental Considerations	0	1	2	3	4	5
Energy Efficiency	0	1	2	3	4	5
Performance / Function	0	1	2	3	4	5
Durability	0	1	2	3	4	5
Maintenance	0	1	2	3	4	5
Production Method	0	1	2	3	4	5
Ergonomics / Style / Appearance	0	1	2	3	4	5
Cost	0	1	2	3	4	5
Sub-Total	0	0	0	6	20	15
Total Score	-	-	-	-	-	41

Table 2.4: *SWOT Analysis for AI-Driven Rack and Pinion Mechanism*

SWOT Analysis	**Strengths**	**Weaknesses**	**Opportunities**	**Threats**
	Active pushing prevents jams.	Requires constant internet connection.	Can easily be updated to treat new types of injuries by changing the API prompt.	If the Gemini API changes or goes offline, the system breaks.
	AI triage removes user cognitive load.	High coding complexity.	–	–

Table 2.5: *Simulation Results, Evaluation, and Conclusion for Design Option 1*

Simulations / Tests Completed	CAD interference testing to ensure the pinion gear teeth mesh perfectly with the rack without binding. Python API latency tests (averaging $3-5$ seconds per request).
Evaluation Against Initial Criteria	Easily passes the 15-second latency threshold (API calls take 5 seconds, physical dispensing takes 3 seconds). Passes the $>90\%$ reliability test because the active thrust of the rack and pinion easily overcomes packaging friction.
Conclusion	The AI-Driven Rack and Pinion design is the definitive preferred option. It actively solves the core problem (the Panic Gap) by providing hands-free diagnosis, and its mechanical design guarantees that flexible soft-packaging will not jam during an emergency.
System Images	 *Figure 2.7: Design Option 1 - Linear Rack and Pinion cartridge mechanism, allowing for active pushing of flexible medical supplies.*
Research Links	Rack and Pinion Kinematics Raspberry Pi to Arduino Serial Communication

2.3.2 Design Option 2

Table 2.6: *Technical Specifications and System Description for Design Option 2*

Design Option 2 - Voice-Activated Rotating Dispenser	
System Description	This alternative design is a hands-free dispenser that uses a local voice recognition module instead of an AI camera. When a user says a wake-word followed by their injury (e.g., “I need a bandage”), the Arduino calculates how many steps to turn a rotating circular carousel. A stepper motor turns the carousel until the correct item lines up with a hole in the floor, dropping the item via gravity.
System Function/s	The system constantly listens for a wake-word. Once activated, it processes the voice command locally (without the internet). An Arduino Mega reads this input and commands a stepper motor to rotate a $3D$-printed drum. Gravity pulls the selected item down the exit chute to the user.
System Components	–
Control Systems	Arduino Mega 2560 (chosen because it has multiple hardware serial ports needed for the voice module).
Sensors	DFRobot Offline Voice Recognition Sensor, and a Hall Effect sensor (magnetic switch) so the carousel knows where its “home” position is when it turns on.
Actuators	NEMA 17 Stepper Motor driven by an $A4988$ motor driver module for precise rotation control.
Mechanical Structures	A rotating cylindrical drum (“Lazy Susan” style) divided into vertical tubes that hold the supplies. It sits on a laser-cut acrylic base with one drop-hole.
Programming Structures	Uses the AccelStepper library for smooth motor movements and standard serial communication to read the voice sensor. Uses `millis()` for timing instead of `delay()` to prevent the code from freezing.
User Interface	Completely hands-free. An RGB LED ring changes color (Blue = Listening, Green = Dispensing) to give the user visual feedback.
Power Sources	$12V$ $5A$ wall plug power supply. The $12V$ goes to the stepper motor driver, and a step-down buck converter drops it to 5V to safely power the Arduino.
System Materials / Constructions	External case made from $6mm$ laser-cut acrylic. The internal rotating drum is $3D$ printed using PLA.
Knowledge Requirements – Difficulty Levels	Level 4 (Advanced). Requires learning how to wire and code stepper motor drivers (setting VREF current limits) and training the voice module.

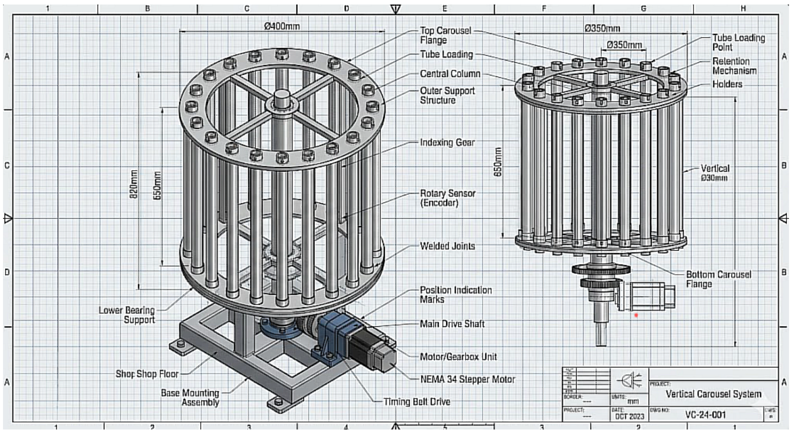
Table 2.7: *Design Option 2 Performance Ratings Against Key Requirements*

Ratings						
User Needs / Requirements	0	1	2	3	4	5
Safety	0	1	2	3	4	5
Environmental Considerations	0	1	2	3	4	5
Energy Efficiency	0	1	2	3	4	5
Performance / Function	0	1	2	3	4	5
Durability	0	1	2	3	4	5
Maintenance	0	1	2	3	4	5
Production Method	0	1	2	3	4	5
Ergonomics / Style / Appearance	0	1	2	3	4	5
Cost	0	1	2	3	4	5
Sub-Total	0	0	6	9	16	0
Total Score	-	-	-	-	-	31

Table 2.8: *SWOT Analysis for Voice-Activated Rotating Carousel*

SWOT Analysis	**Strengths**	**Weaknesses**	**Opportunities**	**Threats**
	Hands-free reduces cross-contamination.	Extremely vulnerable to background noise in a workshop.	Can be programmed with custom school-specific voice commands.	Motor drivers can easily overheat if the current isn't tuned perfectly.
	Very fast processing time because it doesn't need Wi-Fi.	Stepper motors can skip steps and get permanently misaligned.	–	–

Table 2.9: *Simulation Results, Evaluation, and Conclusion for Design Option 2*

Simulations / Tests Completed	Sound level testing in the workshop showed background noise often exceeds 80 decibels when machines are running, which would drown out the user's voice commands.
Evaluation Against Initial Criteria	Fails the > 90% reliability test. Because school workshops are loud, the voice module will frequently misinterpret or miss commands, leaving the injured user frustrated and increasing the "Panic Gap".
Conclusion	The Voice-Activated Rotating Dispenser is a great concept for a quiet nurse's office. However, its fatal flaw is its reliance on voice commands in a loud environment. A visual AI system (camera) is much more reliable in a noisy workshop.
System Images	 *Figure 2.8: Design Option 2 - Rotating drum mechanism relying on stepper motor accuracy and gravity dispensing.*
Research Links	DFRobot Voice Sensor Stepper Motor Guide

2.3.3 Design Option 3

Table 2.10: *Technical Specifications and System Description for Design Option 3*

Design Option 3 - ID-Card Swiped Trapdoor Dispenser	
System Description	This design treats first aid like a secure vending machine to prevent supplies from being stolen. A student must scan their 13.56 MHz RFID ID card to unlock the machine. They then press a touch button for the item they need. An $ESP32$ microcontroller activates a $12V$ solenoid (an electromagnet with a moving pin) that pulls back a trapdoor for half a second, letting one item drop out of a vertical chute.
System Function/s	The system requires RFID input to authenticate the user, followed by a button press. The $ESP32$ processes this, sends an update to a Google Firebase database over Wi-Fi to log who took the item, and then triggers a relay to fire the solenoid, dropping the item via gravity.
System Components	–
Control Systems	$ESP32$ microcontroller (selected because it has built-in Wi-Fi for logging data).
Sensors	$MFRC522$ RFID Reader for scanning cards. $TTP223$ Capacitive Touch Sensors for selecting items (touch sensors are used because physical buttons get jammed with sawdust).
Actuators	Three $12V$ DC Pull-Type Solenoids with return springs.
Mechanical Structures	Three vertical, rectangular 3D-printed chutes. The items are stacked flat inside. The bottom of the chute is blocked by the metal pin of the solenoid. When activated, the pin pulls back, the item drops, and the spring pushes the pin back into place.
Programming Structures	Uses <SPI.h> and <MFRC522.h> libraries for the scanner. Wi-Fi coding to send data to a database.
User Interface	An RFID tap-zone, LED touch buttons, and a 16x2 LCD screen displaying text instructions ("Tap ID", "Select Item").
Power Sources	$12V$ battery charged by a small solar panel (clean energy). A relay module is used so the $3.3V$ $ESP32$ can safely control the high-power $12V$ solenoids. Flyback diodes are required to protect the circuit from voltage spikes.
System Materials / Constructions	Main body folded from sheet metal for extreme durability. Internal chutes $3D$ printed from PETG plastic.
Knowledge Requirements – Difficulty Levels	Level 4 (Advanced). Requires understanding how to safely wire relays, flyback diodes, and high-current inductive loads. Also requires setting up an IoT (Internet of Things) database.

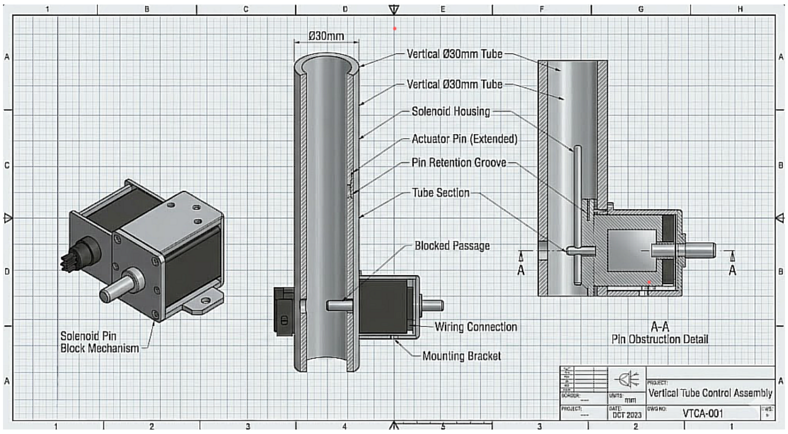
Table 2.11: *Design Option 3 Performance Ratings Against Key Requirements*

Ratings						
User Needs / Requirements	0	1	2	3	4	5
Safety	0	1	2	3	4	5
Environmental Considerations	0	1	2	3	4	5
Energy Efficiency	0	1	2	3	4	5
Performance / Function	0	1	2	3	4	5
Durability	0	1	2	3	4	5
Maintenance	0	1	2	3	4	5
Production Method	0	1	2	3	4	5
Ergonomics / Style / Appearance	0	1	2	3	4	5
Cost	0	1	2	3	4	5
Sub-Total	0	0	6	9	12	5
Total Score	-	-	-	-	-	32

Table 2.12: *SWOT Analysis for ID-Card Swiped Solenoid Trapdoor*

SWOT Analysis	**Strengths**	**Weaknesses**	**Opportunities**	**Threats**
	Logs data to stop students from stealing supplies.	Forces panicked users to find an ID card.	Meets the VCE syllabus requirement for clean energy integration.	Solenoids can easily burn out and melt if the pin gets physically jammed.
	Excellent use of renewable solar energy.	Gravity feeding soft packets causes massive friction jams.	–	–

Table 2.13: *Simulation Results, Evaluation, and Conclusion for Design Option 3*

Simulations / Tests Completed	Friction testing. Soft, crinkly bandage packaging does not slide smoothly against $3D$-printed plastic or metal pins.
Evaluation Against Initial Criteria	Fails the 15-second latency threshold. Finding an ID card, reading a screen, and pressing buttons takes far too much time during a "Panic Gap". It also fails the 90% reliability test because soft packaging will inevitably snag on the solenoid pin mechanism.
Conclusion	The RFID Solenoid system is great for secure tool storage, but awful for first aid. The "Panic Gap" is made worse by locking the supplies behind an ID scan. Furthermore, gravity and solenoids cannot reliably dispense soft packets. An active motorized pushing mechanism (like a rack and pinion) is necessary.
System Images	 *Figure 2.9: Design Option 3 - Vertical gravity-fed magazine utilizing 12V pull-solenoids for actuation.*
Research Links	RFID Scanning Guide Solenoid Guide

2.3.4 Design Option Selection

Design Options Key Strengths vs. Weaknesses

Table 2.14: *Weighted Comparison Matrix of Proposed Design Options*

	Design Option 1	**Design Option 2**	**Design Option 3**
Key Strengths	Highly reliable mechanical push; eliminates user cognitive load entirely.	Fast offline processing; completely touch-free interface.	High security/inventory tracking; uses renewable energy (solar).
Key Weaknesses	Requires internet for AI API; highest programming difficulty.	Highly vulnerable to ambient workshop noise; stepper motors easily misaligned.	Terrible user experience during an emergency; solenoids jam on soft packaging.
Score	41/50	31/50	32/50

Comparison of Design Options

When comparing the three systems against the primary criteria of solving the “Panic Gap”, Design Option 1 is the clear leader. While Option 2 offers great offline processing, its reliance on acoustic voice recognition makes it nearly useless in an 80+ decibel school workshop. Option 3 introduces excellent security and clean energy, but forcing a panicked student to scan an ID card actively worsens the emergency, and its gravity-drop mechanism is inherently incompatible with the flexible nature of bandage packaging. Option 1 balances rapid AI diagnosis with the deterministic, active pushing force of a rack and pinion to guarantee the supply is dispensed reliably.

Therefore Design Option 1 (AI-Driven Rack and Pinion Dispenser) is the preferred design choice.

Justification for Choice of Design Option

Option 1 was selected because it is the only design that meets the Evaluation Criteria defined in Section 1.6. It meets the < 15 second response latency by using the Gemini AI API, circumventing the slow physical menu navigation seen in Option 3. Most importantly, it satisfies the $> 90\%$ dispensing reliability criteria. The “Factors influencing design” prioritize user safety and function; because soft medical packaging inevitably jams in passive gravity systems (Options 2 and 3), the active linear thrust provided by Option 1’s rack and pinion is mandatory to ensure the user receives their first aid supply without failure.

Development of Selected Option

Table 2.15: *Detailed Development Specifications for the Preferred Design Option*

Choice of Materials	6mm Plywood (Pine) for the external chassis due to its high tensile strength and acoustic dampening. PLA (Polylactic Acid) bioplastic for all internal mechanical parts due to its low warping during $3D$ printing.
Choice of Power / Energy Source	$240V$ AC to $5V$ DC ($3A$) mains wall adapter. Batteries were rejected to prevent the system from dying mid-operation, ensuring the Pi and servos have a constant, reliable current draw.
Choice of Controllers	Raspberry Pi 4 Model B (High-level AI & Vision Processing) and Arduino Uno $R3$ (Low-level Actuator Control).
Choice of Sensors	USB Microphone (Audio input), Raspberry Pi Camera Module $V3$ (Visual input), and SPDT Micro Limit Switches (Mechanical feedback).
Choice of Actuators	FS90R Continuous Rotation Servos. Selected over standard servos to allow multiple rotations, achieving the necessary $110mm$ linear travel distance.
Choice of Mechanical Structures	A $3D$-printed rack and pinion mechanism pushing a wedge-shaped sled. A 45-degree plywood ramp at the front of the device to deliver the item to the user.
Choice of Production Methods	Fused Deposition Modeling (FDM) 3D Printing for gears and cartridges. Laser cutting for the plywood chassis to ensure perfectly square joints and clean screen bezels.
Choice of Programming Structures	Python (for Gemini API HTTPS requests, JSON parsing, and UART serial transmission). C++ (for Arduino serial reading and servo PWM control).
Functionality Needs / Requirements	Must process the user’s request and dispense the correct item (Bandage, Gauze, or Wipe) in under 15 seconds with a $> 90\%$ mechanical reliability rate.

Environmental Considerations	Plywood is a renewable timber resource. PLA is derived from cornstarch and is biodegradable under industrial composting conditions.
Safety Considerations	All high-voltage ($240V$) components are isolated outside the box in a certified wall-adapter. The internal chassis runs entirely on safe $5V$ DC. No sharp edges are exposed.
Cost	Approximately $\$320$ for materials and components.
Dimensions / Weight	$400mm\ (W) \times 300mm\ (D) \times 300mm\ (H)$. Estimated weight: 2.5 kg.
Maintenance	A hinged rear access door allows the school nurse or teacher to reload the cartridges without interacting with or removing the delicate electronic components.
Ergonomics / Style / Appearance	A clean, angled desktop kiosk. The 7-inch touchscreen is angled at 45 degrees for optimal viewing by a standing user.
Standards & Regulations	Compliance with school electrical safety standards (using RCM-certified power supplies) and adhering to basic First Aid supply storage guidelines (keeping supplies in original sterile packaging).

2.4 Preferred Design Option

2.4.1 Feasibility Analysis

Table 2.16: *Final Feasibility Analysis and Conceptual Inheritance*

Design Option	Feasible?	Justification for Feasibility	Ideas/Parts Taken from Idea
Option 1: AI Camera + Rack & Pinion	Highly Feasible	Fits within budget using the Raspberry Pi 4 and basic Arduino. The 3D-printed rack and pinion can be iterated easily on school printers. The Gemini AI API provides robust software without needing to train custom models.	The core dual-microcontroller architecture and the linear pushing mechanism.
Option 2: Voice-Activated Carousel	Partially Feasible	Easy to construct and cheap. However, achieving reliable voice recognition in an $80+$ decibel workshop is practically unfeasible for a Year 12 project.	The concept of hands-free interaction to prevent cross-contamination.
Option 3: RFID Swiped Trapdoor	Low Feasibility	Setting up an online database, solar charge controllers, and tuning high-current solenoids will likely exceed the time limit. Furthermore, solenoids jamming on soft bandages makes it mechanically flawed.	The concept of modular, $3D$ printed cartridges for holding the supplies.

2.4.2 Preferred Design

The chosen system is Design Option 1: The AI-Driven Rack and Pinion Dispenser (V.I.S.O.R). This design combines the Raspberry Pi 4 (using the Gemini AI API for visual/audio diagnosis) with an Arduino Uno (to control continuous rotation servos). The mechanical system utilizes a $3D$-printed rack and pinion to actively push the required first aid supply out of a modular cartridge.

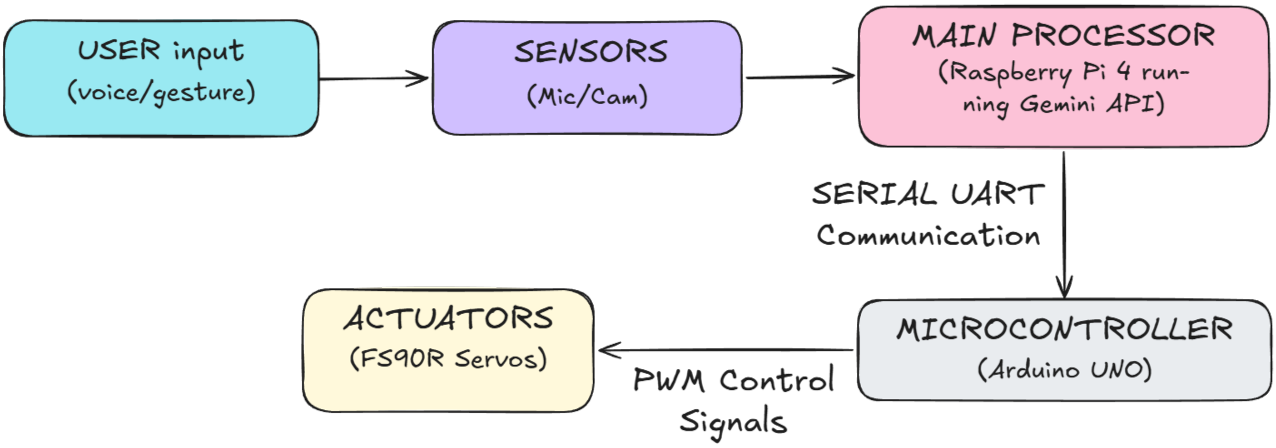


Figure 2.10: The finalized system architecture for the preferred V.I.S.O.R design, separating high-level AI logic from low-level motor control.

2.4.3 Justification for Preferred Design

Design Option 1 was selected because it is the only system that definitively solves the “Panic Gap” outlined in the original Design Brief (Section 1.1) while maintaining a $> 90\%$ dispensing reliability.

Meeting the Requirements: As established in the Design Brief, the system must reduce cognitive load during an emergency. Option 3 (RFID/Buttons) actively worsened the Panic Gap by forcing the user to find an ID card and navigate a menu. Option 1 allows the user to simply state their problem or show the injury to the camera, fulfilling the hands-free, low-stress requirement. Furthermore, the 15-second latency criteria can be met by utilizing the Gemini API, which processes requests much faster than the manual menu navigation of Option 3.

Mechanical Reliability: First aid supplies, such as bandages and gauze, are soft and packaged in flexible wrappers. Option 2 and Option 3 relied on gravity to drop these items. Research and testing indicated that soft packaging easily snags on 3D printed layers or solenoid pins, causing jams. Option 1 was chosen because a rack and pinion mechanism applies active, linear force to the back of the packet, forcibly pushing it out of the cartridge. This is the only mechanical method evaluated that can guarantee the $> 90\%$ reliability requirement.

Rejection of Alternatives:

- Design Option 2 was rejected due to its environmental vulnerability. The acoustic noise of a standard school workshop would render the offline voice sensor completely useless, failing the functional requirement.
- Design Option 3 was rejected due to its poor emergency user experience and high likelihood of mechanical jamming, violating the primary user safety constraint.

2.5 Integrated System Modelling

2.5.1 Design Overview

The V.I.S.O.R system is designed as a desktop kiosk. The exterior is a 6mm plywood chassis with an angled roof housing a 7-inch HDMI touchscreen. Internally, the space is divided into two zones. The front zone contains three identical, $3D$-printed vertical cartridges holding Bandages, Gauze, and Alcohol Pads. The rear zone contains the electronic "brain" (Raspberry Pi and Arduino) mounted to the back wall, and three $FS90R$ servos mounted directly behind each cartridge. When activated, the servos push a rack through a hole in the back of the cartridge, forcing the medical supply out of a $2.5mm$ front slit. The item hits a 45-degree internal wooden ramp and slides out to the user.

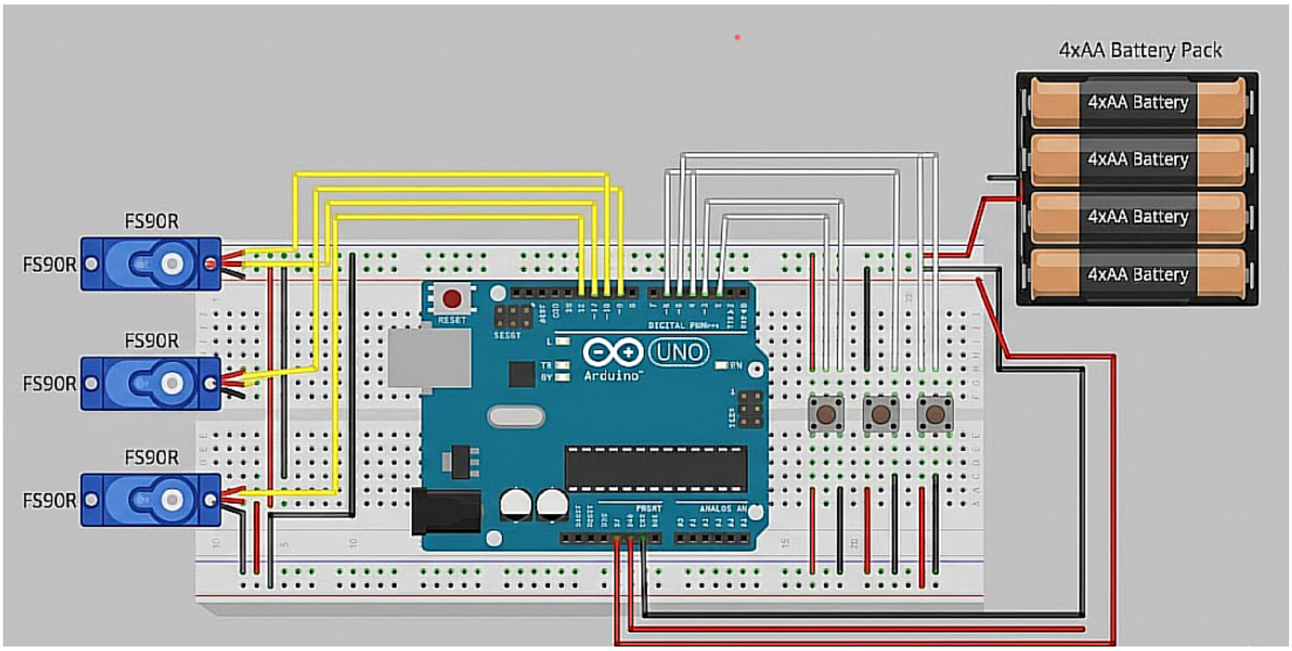


Figure 2.11: Basic electronic schematic demonstrating the low-voltage logic and PWM control wiring for the electromechanical subsystems.

2.5.2 Mechanical Structure

The core mechanical structure relies on a Rack and Pinion system to convert rotational kinetic energy into linear thrust.

- **The Pinion (Gear):** Attached directly to the spline of the $FS90R$ continuous rotation servo.
- **The Rack (Track):** A flat bar with corresponding gear teeth, attached to a "pusher sled" inside the cartridge. As the servo rotates, the pinion teeth engage the rack, pushing the sled forward. Because medical supplies are soft and flexible, a simple gravity drop would cause them to jam. The wedge-shaped sled applies active, horizontal force directly to the back of the packaging, pushing it cleanly through the tight front slit. Once the item drops, the servo reverses. A mechanical limit switch mounted at the rear of the cartridge is physically pressed by the returning rack, closing the circuit and telling the Arduino to stop the motor exactly at its "home" position.

2.5.3 Control System

IPO Flowchart

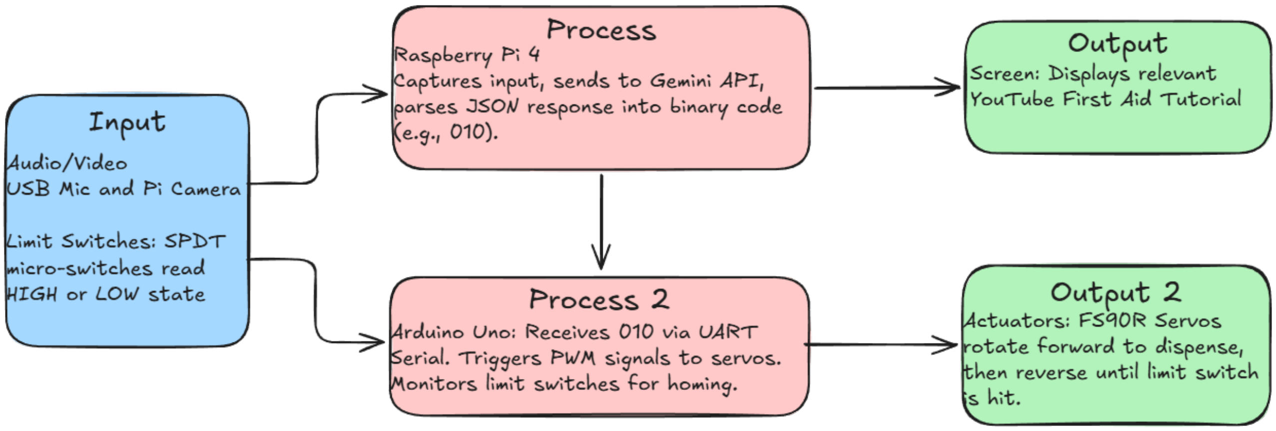


Figure 2.12: IPO Flowchart for the V.I.S.O.R System

Input Systems (Sensors)

1. **USB Microphone & Pi Camera** $V3$: Used to capture the user's distress signal (natural language or visual injury). They feed unstructured data directly into the Raspberry Pi.
2. **Micro Limit Switches (SPDT)**: Mechanical sensors placed at the rear of the rack track. When the rack is fully retracted, it presses the switch arm, closing the circuit to the Arduino's digital read pins. This acts as essential closed-loop mechanical feedback to prevent the servo from stripping the gears.

Table 2.17: *Technical Subsystem Overview (Sensors, Actuators, and Control)*

Subsystem	Description
Electrical	$5V$ $3A$ power distribution. Logic level shifting (if required) between Pi ($3.3V$) and Arduino ($5V$).
Mechanical	Rack and pinion gears, pusher sleds, and the 45-degree delivery ramp.
Control Systems (Programmed)	Python (API and UI handling) and C++ (Motor timing and interrupt polling).
Communication	Wi-Fi (Pi to Gemini API) and UART Serial (Pi to Arduino via USB/TX-RX).
Sensors	Camera, Microphone, and Mechanical Limit Switches.
Actuators	Continuous Rotation Servos generating rotational torque.

Process (Microcontroller)

The system uses a Dual-Microcontroller Architecture:

1. **Raspberry Pi** 4 **(High-Level)**: Processes the heavy audio/video data, connects to the internet via Wi-Fi, and handles the Gemini API. It translates complex API text into a simple 3-digit binary string (e.g., "100" for Bandage, "010" for Gauze).
2. **Arduino Uno** $R3$ **(Low-Level)**: Dedicated entirely to hardware actuation. It constantly listens to the Serial port.

Pseudocode:

```
START
    Pi listens for User Audio
    Pi sends Audio to Gemini API
    Pi receives response -> extracts binary "010" and YouTube URL
    Pi opens Chromium browser with YouTube URL
    Pi sends "010" to Arduino via Serial (9600 baud)

    Arduino reads "010"
    Arduino writes PWM to Servo 2 (Gauze) -> Rotate Forward
    Delay (Time taken to push rack 110mm)
    Arduino writes PWM to Servo 2 -> Rotate Reverse

    WHILE (Limit Switch 2 == NOT PRESSED):
        Keep Rotating Reverse

    Arduino writes PWM to Servo 2 -> STOP
END
```

Listing 2.1: *V.I.S.O.R. System Logic Pseudocode*

Outputs (Actuators)

1. $FS90R$ **Continuous Rotation Servos**: These convert electrical PWM signals into rotational mechanical energy. They operate on $4.8V - 6V$ and provide approximately 1.5 kg-cm of torque, which is more than sufficient to overcome the sliding friction of the PLA rack and a 50-gram stack of bandages.
2. 7**-inch HDMI Touchscreen**: Acts as the visual actuator, providing the user with immediate, calming, and instructional visual feedback (the YouTube video) to ensure the dispensed item is applied correctly.

3 Developing a Work Plan

3.1 Key Deliverables & Milestones

To ensure the V.I.S.O.R project stays on track within the strict Year 12 timeline, the development has been broken down into three major functional milestones:

1. **Proof of Concept (PoC)**: The foundational step. This involves establishing successful UART serial communication between the Raspberry Pi and the Arduino. The milestone is reached when typing a manual "100" or "010" into the Pi's terminal successfully triggers the Arduino to spin a single FS90R continuous rotation servo and stop when a limit switch is manually pressed.
2. **Minimum Viable Product (MVP)**: The core functionality test. This milestone integrates the mechanical and software systems for a single item. The Pi must successfully record a voice prompt via the USB microphone, query the Gemini AI API, receive the correct binary string, and actuate the servo. The servo must successfully push a single bandage out of a 3D-printed prototype cartridge using the rack and pinion mechanism.
3. **Final System Integration**: The completed prototype. This includes manufacturing the laser-cut plywood chassis, integrating all three cartridges (Bandages, Gauze, Alcohol Pads), mounting the 7-inch HDMI touchscreen to display the YouTube instructional videos, and passing the evaluation criteria metrics ($< 15s$ latency, > 90% reliability).

3.2 Project Timeline

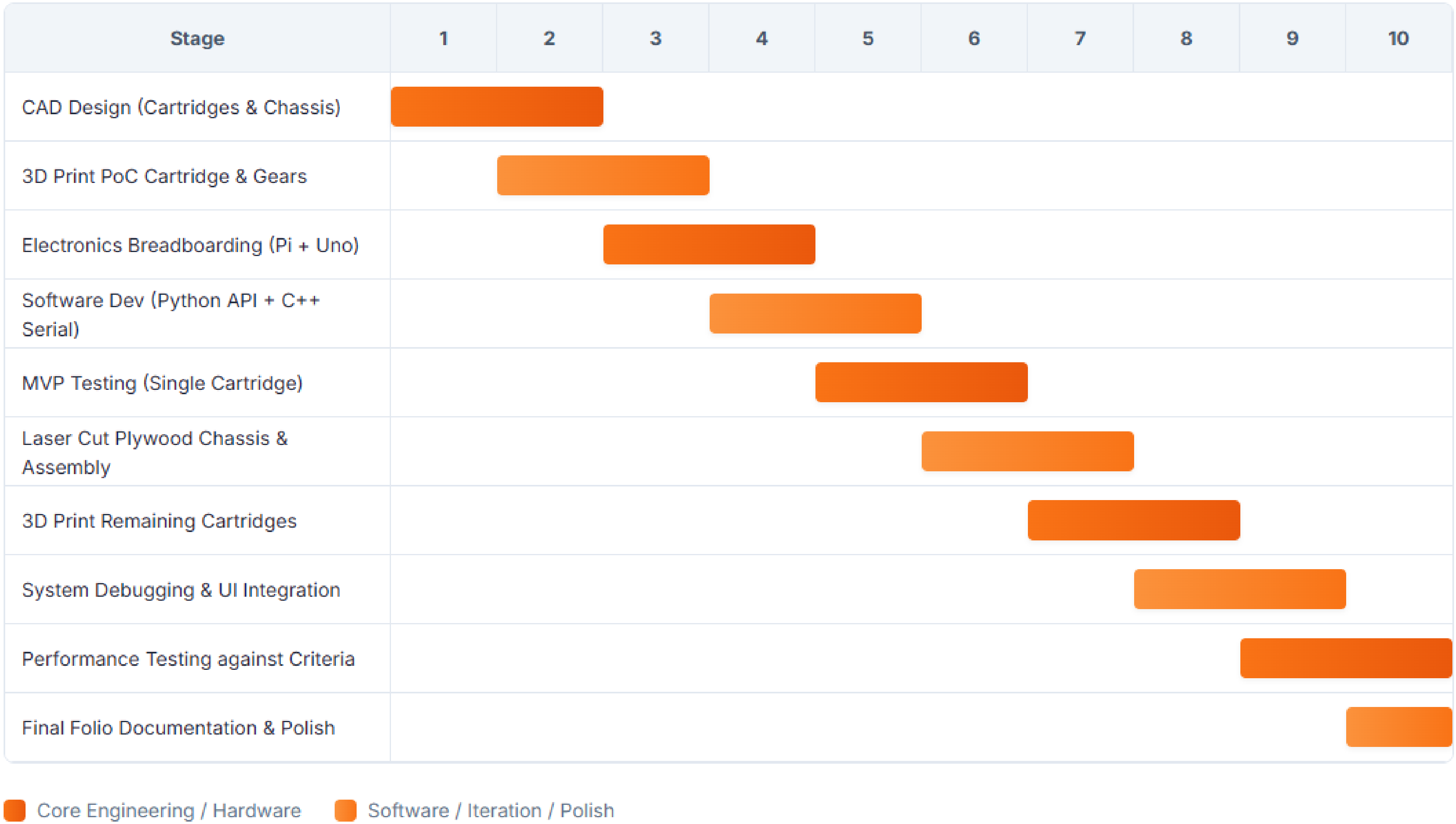


Figure 3.1: *Visual Gantt chart depicting the critical path and parallel dependencies of the CAD, software, and electronic production phases.*

Production Work Plan

Table 3.1: *Detailed Production Work Plan, Equipment, and Justification*

Stage	Stage in Engineering Process	Process	Equipment	Safety/Risk	Justification
1	CAD Modelling & Slicing (Designing and Modelling)	Parametric modelling of tolerances; converting STL files to G-Code.	PC with Fusion 360 and Prusa Slicer.	Screen fatigue, ergonomic strain.	Ensures the $1.5mm$ tolerances between the rack and pinion are mathematically perfect before wasting physical material.
2	3D Printing Mechanics (Producing/ Manufacturing)	Fused Deposition Modelling (FDM) of PLA plastic.	School 3D Printers (Prusa)	Burn hazard from 200 Degrees Celsius nozzle; pinch points in belts	Complex involute gear teeth for the rack and pinion cannot be accurately manufactured by hand; $3D$ printing is mandatory.
3	Chassis Fabrication (Producing/ Manufacturing)	Vectoring DXF files and laser cutting the 6mm pine plywood.	CO2 Laser Cutter.	Fire Hazard; laser scatter eye damage	Provides the structural rigidity required to house the 7-inch screen and $240V$ power adapter securely.
4	Electronics Assembly (Integrating & Assembling)	Stripping wires, crimping jumper cables, and soldering the limit switch harness.	Soldering Iron, Wire Strippers, Multimeter.	Thermal burns; toxic flux fume inhalation	Ensures reliable electrical connections, preventing voltage drops between the Arduino and the high-current servos.
5	System Testing & Calibration (Testing and Evaluating)	Running Python/C++ code, tuning PWM delay times for $110mm$ rack travel.	PC, completed V.I.S.O.R prototype.	$240V$ mains shock hazard from wall plug.	Required to tune the software to hit the strict < 15-second latency and > 90% reliability evaluation criteria.

Actual Project Timeline

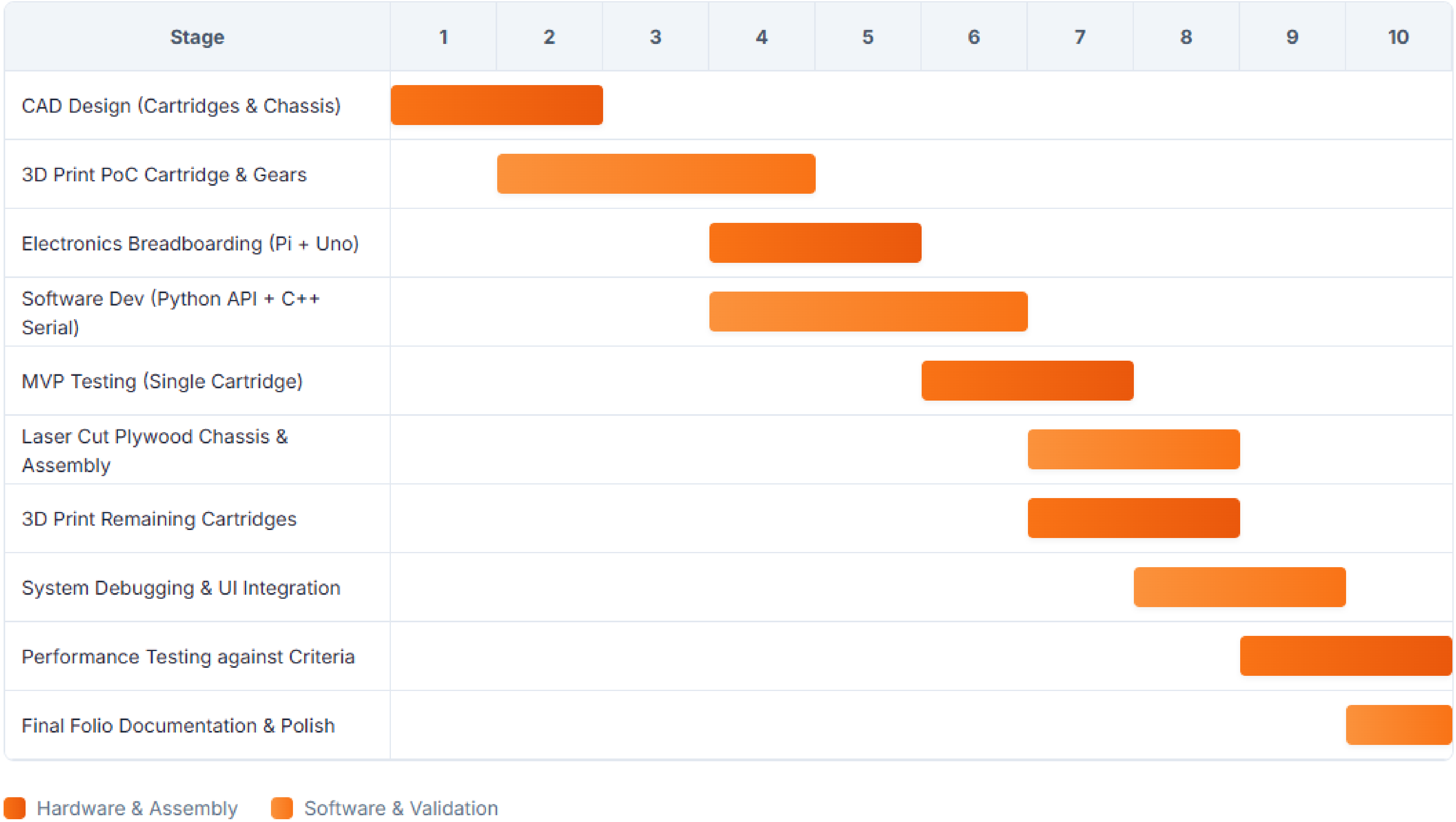


Figure 3.2: *Visual Gantt chart depicting the actual project timeline*

3.3 Project Budget

The total estimated budget for the V.I.S.O.R system is $\$322.25$ AUD. Because this project integrates advanced computing (AI and Vision) alongside mechanical actuation, the bulk of the budget is consumed by the ”brain” components—specifically the Raspberry Pi 4 ($\$110.00$) and the 7-inch HDMI Screen ($\$95.00$). To ensure the project remains financially viable, structural materials have been heavily optimized. The school’s supply of PLA filament will be utilized for the internal mechanics, reducing external polymer costs to zero, and the primary chassis utilizes highly affordable 6mm Pine Plywood ($\$15.00$). All electronic components will be purchased in a single bulk order from Core Electronics and JayCar at the beginning of Term 2. Ordering early prevents internet shipping delays from pushing the production Gantt chart off schedule, ensuring parts arrive in time for the Breadboarding phase in Week 3.

3.4 Required Materials

All items will be purchased in bulk orders from JayCar Electronics and Core Electronics at the beginning of Term 2 to avoid shipping delays impacting the Gantt chart timeline. The PLA filament is provided by the TGC technology department.

Table 3.2: *Itemised Project Budget and Component Procurement List*

Item	Supplier	Quantity	Unit Price	Total Price
Raspberry Pi 4 Model B (4GB)	Core Electronics	1	$\$110.00$	$\$110.00$
Arduino Uno $R3$	JayCar	1	$\$39.95$	$\$39.95$
7-inch HDMI Touchscreen	Core Electronics	1	$\$95.00$	$\$95.00$
$FS90R$ Continuous Rotation Servo	JayCar	3	$\$9.95$	$\$29.85$
Micro Limit Switches (SPDT)	JayCar	3	$\$2.50$	$\$7.50$
$5V$ $3A$ Mains Power Supply	JayCar	1	$\$19.95$	$\$19.95$
$6mm$ Plywood Sheet (1200x600mm)	Bunnings	1	$\$15.00$	$\$15.00$
PLA $3D$ Printer Filament ($1kg$)	School Supply	1	$\$0.00$	$\$0.00$
Assorted $M3$ Screws and Nuts	Bunnings	1	$\$5.00$	$\$5.00$
USB Microphone Mini Dongle	Amazon	1	$\$10.00$	$\$10.00$
Grand Total	–	–	–	$\$322.25$

3.5 Technical Resource Analysis

3.5.1 Materials

The preferred option makes use of 6mm Pine Plywood for the main structural chassis. Plywood has high tensile strength, provides excellent acoustic insulation for internal motors, and is highly sustainable (a renewable resource). It is easily machined using the school's laser cutter and provides excellent structural purchase for $M3$ mounting screws. For the internal dispensing cartridges, PLA (Polylactic Acid) plastic was selected. PLA is a biodegradable bioplastic that offers extremely low warping during $3D$ printing, which is crucial for maintaining the precise $1.5mm$ sliding tolerances required for the rack and pinion gears to mesh without binding.

3.5.2 Components

A Raspberry Pi 4 Model B was selected as the primary brain because its $4GB$ of RAM and Linux OS are necessary to process HTTP requests to the Gemini AI API and drive the 7-inch HDMI display. However, because Raspberry Pi OS struggles with real-time hardware timing, an Arduino Uno $R3$ was selected as a secondary subsystem solely to generate the precise Pulse Width Modulation (PWM) signals required by the motors. The $FS90R$ Continuous Servos were chosen over standard 180-degree servos because the rack needs to travel $110mm$; a continuous servo can spin multiple full rotations to push the rack the required distance.

3.5.3 Processes and Tools

FDM 3D Printing will be used to manufacture the rack and pinion gears and the cartridges. This process was selected over traditional woodworking because it allows for rapid prototyping of complex involute gear teeth that cannot be cut by hand. A Laser Cutter will be utilized to cut the plywood chassis; this process ensures perfectly straight edges and precise cut-outs for the screen bezel, drastically improving the final ergonomic appearance and safety of the device compared to using a hand-held jigsaw. Assembly will primarily utilize non-permanent fasteners ($M3$ machine screws) to allow for easy disassembly and maintenance.

Table 3.3: *Occupational Health and Safety (OHS) Risk Assessment and Control Matrix*

Equipment or Task/Process	Hazard Identification	Risk Assessment			Risk Control Measures		
		Consequence	**Likelihood**	**Risk Rating**	**Engineering**	**Procedures**	**PPE**
Laser Cutter (Plywood)	Fire hazard if wood ignites; eye damage from laser scatter.	Major	Rare	High	Lid safety interlocks engaged; exhaust fan on.	Never leave cutter unattended while firing.	None required (interlock prevents exposure).
FDM $3D$ Printing	Contact burns from $200°C$ nozzle; pinch hazard in moving belts.	Minor	Possible	Medium	Printer enclosed in safety cabinet	Wait for bed to cool before removing PLA parts.	None required.
Soldering Iron (Electronics)	Thermal burns to skin; Inhalation of toxic flux fumes.	Minor	Likely	High	Fume extractor running	Solder in designated safe zones, return iron to stand.	Safety glasses.
Utility Knife (Post-processing)	Deep lacerations to fingers when removing $3D$ print supports.	Moderate	Possible	High	Use cutting mat.	Always cut away from the body; use pliers for stubborn supports.	Cut-resistant gloves.
Power Tools (Cordless Drill)	Puncture wounds from drill bit; hair/clothing getting caught.	Moderate	Possible	High	Chuck tightened securely.	Clamp plywood firmly to bench before drilling; tie hair back.	Safety glasses.
$240V$ Mains Power	Fatal electric shock when plugging in the system.	Severe	Rare	High	Use ONLY pre-certified, double-insulated commercial $5V$ wall adapters. No bare $240V$ wiring.	Ensure hands are dry.	Rubber-soled shoes.